Dependent Gaussian Plaquette Polymers on a Four-Dimensional Gauge Block

Shared-link and Bianchi dependence without an independence hypothesis

20 July 2026

Abstract

The independent-defect benchmark used in an ultraviolet Yang–Mills calibration does not represent shared links or Bianchi constraints. We replace it by the exact linearized gauge quotient on the open side-two four-dimensional block. Its curvature is Gaussian with covariance $tP \otimes I_3$, where P is the orthogonal projection onto the 136-dimensional scalar exact-curvature subspace. Projection contraction yields, for every set of k plaquettes, a simultaneous radial-tail estimate $\exp[-kJ_3(\rho^2/t)]$, with $J_3(s) = \frac{1}{2}(s - 3 - 3\log(s/3))$. Combining this joint estimate with the exact degree-18 adjacency graph gives a subcritical rooted connected-set sum at the frozen ultraviolet witness, without assuming independence. The result is exact for the linearized Gaussian model; comparison with the nonlinear SU(2) measure remains open.

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1 Gauge quotient in curvature coordinates

Let $d_1 : C^1 \to C^2$ be the scalar link-to-plaquette coboundary. On the contractible block, $\ker d_1$ consists of gauge directions. Quotienting that kernel and mapping to curvature identifies the physical scalar space with $\operatorname{im} d_1$, of dimension 136.

For three color components, the Gaussian action is

$$S(F) = \frac{1}{2t}\|F\|^2, \qquad F \in \operatorname{im} d_1 \otimes \mathbb{R}^3.$$

The Jacobian of the linear quotient-to-curvature map is constant. Therefore F is the isotropic Gaussian on this subspace. In the ambient $\mathbb{R}^{216} \otimes \mathbb{R}^3$ coordinates,

$$\operatorname{Cov}(F) = tP \otimes I_3, \tag{1}$$

where P is the orthogonal projection onto $\operatorname{im} d_1$.

The executable matrix has idempotence error 6.33×10^{-15}, eigenvalues in numerical interval $[-7.04 \times 10^{-16}, 1 + 6.7 \times 10^{-15}]$, and rank 136. These computations certify the declared finite matrix; the abstract property $0 \le P \le I$ follows exactly from orthogonal projection.

2 A joint radial tail with dependence retained

For a plaquette set S of size k, let $F_S \in \mathbb{R}^{3k}$ be the coordinate restriction. Its covariance is a principal compression of (1) and hence is bounded above by tI_{3k}.

Theorem 2.1 (Dependent simultaneous-defect estimate)**.** *Let $s = \rho^2/t$. If $s > 3$, then for every S with $|S| = k$,*

$$\mathbb{P}\{\|F_p\| \ge \rho \text{ for all } p \in S\} \le \exp[-kJ_3(s)], \qquad J_3(s) = \frac{1}{2}\left(s - 3 - 3\log\frac{s}{3}\right). \tag{2}$$

No independence assumption is made.

Proof. The event implies $\|F_S\|^2 \ge k\rho^2$. For $0 < \lambda < (2t)^{-1}$, covariance contraction and the Gaussian determinant formula give

$$\mathbb{E}e^{\lambda\|F_S\|^2} \le (1 - 2t\lambda)^{-3k/2}.$$

Markov's inequality yields

$$\mathbb{P}\{\|F_S\|^2 \ge k\rho^2\} \le e^{-\lambda k\rho^2}(1 - 2t\lambda)^{-3k/2}.$$

The optimizer is $2t\lambda = 1 - 3/s$, which is admissible for $s > 3$. Substitution proves (2). □

3 Rooted connected-set theorem

Use plaquette adjacency by a shared link. Its exact maximum degree is $\Delta = 18$. A canonical spanning-tree traversal bounds the number of connected sets of size k containing a fixed root by $\Delta^{2(k-1)}$.

Corollary 3.1 (Dependent Gaussian connected-event sum)**.** *Set $q_G = e^{-J_3(s)}$. If*

$$r = e^{\alpha}\Delta^2 q_G < 1, \tag{3}$$

then

$$\sup_p \sum_{\substack{X \ni p \\ X \text{ connected}}} e^{\alpha|X|}\mathbb{P}\{X \text{ is entirely bad}\} \le \frac{e^{\alpha} q_G}{1 - r}. \tag{4}$$

Proof. Apply (2) for each connected X, insert the connected-set count, and sum the geometric series from $|X| = 1$. □

4 Ultraviolet witness

For $t = 3 \times 10^{-6}$ and $\rho = 0.0082$,

quantity	value
$s = \rho^2/t$	22.4133333
$J_3(s)$	6.69010107
q_G	0.00124315712
$e^{0.1}18^2 q_G$	0.445143955
rooted sum upper	0.00247613973
maximum certified t	$3.27496193 \times 10^{-6}$

The maximum heat time is obtained by monotone bisection of $J_3(\rho^2/t) = \alpha + 2\log 18$. The witness is strict, with heat-time slack 2.74962×10^{-7}.

5 Frontier audit

This theorem shows that shared-link and linear Bianchi dependence do not by themselves close the ultraviolet window. It does not control nonlinear BCH remainders, Haar densities, incoherent collars, noncentral topological data, or repeated RG. Equation (4) is a connected bad-event envelope; it is not by itself a Mayer or Ursell activity expansion. The next bridge must compare the nonlinear measure with this Gaussian law on a coherent collar and separately charge collar failure. The result is C2 for the linearized Gaussian gauge model.